Employee Attrition Prediction Using Machine Learning

Completed Major Project Report | Evaluation, Results, Limitations and Conclusion

Results at a glance

Metric	Reported result
Accuracy	73.1%
Precision	35.2%
Recall	80.9% (38/47 leavers detected)
F1-score	0.490
ROC-AUC	0.803
Average precision / PR-AUC	0.561
Decision threshold	0.39
Test set	294 employees; 47 leavers

Abstract

Employee attrition can increase recruitment costs, disrupt team knowledge and affect delivery continuity. This project presents an educational machine-learning workflow for analyzing historical attrition labels, evaluating a classifier and demonstrating results through a Streamlit dashboard.

The final presentation reports a recall-focused Logistic Regression model on a held-out test set of 294 employees. At threshold 0.39, reported metrics are 73.1% accuracy, 35.2% precision, 80.9% recall, 0.490 F1-score, 0.803 ROC-AUC and 0.561 average precision. The confusion matrix below is reconstructed from these reported figures and should be checked against the application output.

The IBM HR Analytics dataset is an educational sample without time-stamped snapshots or a six-month outcome window. The model predicts the historical Attrition label; it is not a validated six-month forecasting system.

1. Introduction

Employee attrition can increase replacement costs, remove institutional knowledge and create workload pressure for remaining employees. Workforce analytics can summarize patterns in historical departures and help teams consider voluntary support initiatives. Model outputs are uncertain signals and should not replace human judgement or employee dialogue.

2. Problem Statement and Objectives

The goal is to build an end-to-end prototype that learns from employee attributes and historical Attrition labels, evaluates minority-class performance, communicates results to non-technical stakeholders and exposes an interactive dashboard.

Objectives are to explore data quality, preprocess numeric and categorical features reproducibly, account for class imbalance, evaluate a classifier, present results through Streamlit, and communicate limitations, fairness concerns and responsible-use boundaries.

3. Dataset Description

The project uses the commonly distributed IBM HR Analytics Employee Attrition & Performance dataset from Kaggle. The standard version contains 1,470 rows and 35 columns, including 237 Attrition=Yes and 1,233 Attrition=No labels (about 16.1% positive). Features include age, department, job role, job level, income, overtime, satisfaction, work-life balance and tenure.

The dataset is synthetic/educational and should not be described as current data from a real employer. Confirm the counts against the exact CSV used for the final run.

4. Methodology

Data validation: check for the Attrition target, valid Yes/No labels, missing values and expected column names.

Feature handling: separate the target from predictors. Remove identifier-like or constant columns such as EmployeeNumber, EmployeeCount, Over18 and StandardHours where present. Numeric features use median imputation and standardization; categorical features use most-frequent imputation and one-hot encoding.

Train/test split: use a stratified 75/25 split with a fixed random seed. Fit preprocessing inside the training pipeline so the test set does not influence imputation, scaling or encoding.

Model and threshold: the final presentation reports recall-focused Logistic Regression at threshold 0.39, selected using cross-validated predictions on training data rather than the test set. Class weighting gives more attention to the minority class. The provided final presentation does not include a complete comparable Random Forest results table, so no claim is made that Logistic Regression outperforms Random Forest.

Metrics: precision is the share of flagged employees with a positive historical label; recall is the share of positive labels identified; F1 is the harmonic mean of precision and recall; ROC-AUC measures ranking across thresholds; average precision summarizes the precision-recall curve; accuracy is secondary because the target is imbalanced.

5. Results and Discussion

The metrics below are transcribed from the final presentation and have not been independently rerun while preparing this report. The presentation identifies 294 test employees and 47 leavers.

6. Confusion Matrix

At threshold 0.39, the presentation reports that 38 of 47 leavers were detected. Combining this with reported precision of 35.2% and the 294-person test set gives the reconstructed confusion matrix. It is consistent with reported accuracy: (38 + 177) / 294 = 73.1%. Verify the counts against the exact application output before submission.

Actual / predicted	Predicted Left (Yes)	Predicted Stayed (No)
Actual Left (Yes)	38 — True Positive	9 — False Negative
Actual Stayed (No)	70 — False Positive	177 — True Negative

Counts: TP=38, FN=9, FP=70, TN=177. Accuracy=(38+177)/294=73.1%. Verify this reconstructed matrix against the exact app output.

7. Interpretation

Recall of 80.9% means 38 of the 47 employees labelled as leavers in the test set were detected; 9 were missed. Precision of 35.2% means roughly one in three flagged employees had a positive historical label, so false alarms are common. This is a recall-first trade-off: more historical leavers are caught, but more employees are flagged unnecessarily.

Accuracy of 73.1% should not be interpreted alone. ROC-AUC of 0.803 suggests useful ranking discrimination on this holdout, but it does not prove calibrated individual probabilities or performance on future employees. Describe outputs as risk scores, not literal probabilities such as “70% chance of leaving.”

8. Limitations

Educational data: the IBM sample is synthetic and does not represent live data from a specific employer.

Small positive test class: only 47 leavers appear in the test set, so recall estimates are uncertain and may change with another split or dataset.

False positives: precision is 35.2%; many flagged cases are not positive in the held-out labels.

No six-month horizon: the dataset has no observation dates, longitudinal snapshots or defined six-month follow-up label. The model cannot establish when a departure will occur and is not a validated six-month probability model.

Fairness and sensitive features: age, gender and marital status are included in the described model features, but no subgroup fairness audit is reported. A real pilot requires privacy/legal review, subgroup error-rate checks and calibration evaluation.

Dashboard behavior: presentation notes state that CSV upload may update charts while predictions continue to use the original model and require the expected IBM column layout. Explain and test this behavior before a live demo.

Reproducibility: this report uses metrics from the presentation rather than an independent model run. A full Random Forest comparison is not available in the final presentation, and results may vary with dataset version, library versions, split and threshold.

9. Ethics and Responsible Use

Treat scores as uncertain signals for voluntary support and organizational improvement. Do not use them to fire, discipline, rank employees, deny promotion or apply punitive monitoring. Protect privacy, restrict access, document the purpose, include human oversight and consult affected stakeholders. Associations are not proof of causation.

10. Conclusion and Future Work

This project demonstrates an educational workflow for attrition-label analysis, including preprocessing, class-weighted classification, evaluation and a Streamlit interface. On the reported holdout, the recall-focused Logistic Regression model achieved 80.9% recall, while precision was 35.2%, highlighting the trade-off between detecting historical leavers and generating false alarms.

The system is not a validated operational early-warning or six-month forecasting tool. Future work should verify the confusion matrix from the exact model run, compare models on the same split or repeated cross-validation, assess calibration and subgroup errors, and test the dashboard end to end. Genuine six-month prediction requires de-identified longitudinal snapshots, a defined index date and outcome window, leakage controls and temporal evaluation.

References

Kaggle — IBM HR Analytics Employee Attrition & Performance:
https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Streamlit Community Cloud documentation: https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app

Scikit-learn documentation: https://scikit-learn.org/stable/